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SSGMRN: Spatial-spectral global modeling retentive network for hyperspectral image denoising

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ABSTRACT

Noise pollution is an inevitable issue in hyperspectral image (HSI) acquisition. It significantly limits the effectiveness of subsequent HSI applications. Deep learning-based methods have shown tremendous potential for HSI denoising. However, traditional convolutional or Transformer-based approaches are limited by computational efficiency and lack clear spatial and spectral priors. To address this issue, we propose the spatial-spectral global modeling retentive network (SSGMRN), which introduces distance-dependent spatial and spectral priors for HSI denoising. SSGMRN consists of multiple spectral and spatial retention blocks, which enhance the extraction of global spatial and spectral features through explicit spatial and spectral decay mechanisms. In the spatial domain, a two-dimensional bidirectional spatial decay matrix is designed based on the Manhattan distance between pixels. This decay matrix is combined with an attention decomposition mechanism, enabling efficient modeling of spatial information in HSI with reduced computational complexity. In the spectral domain, the proposed decay matrix effectively assists in modeling spectral information, enhancing the network's ability to learn spectral features across different bands in HSI. Furthermore, the guidance provided by the decay matrices allows the model to more effectively preserve the variations in spatial and spectral information with respect to distance. Extensive simulation and real experiments demonstrate that, compared with state-of-the-art methods, SSGMRN achieves superior performance in HSI denoising. Codes are available at <https://github.com/Featherrain/SSGMRN.git>.

1. Introduction

Hyperspectral image (HSI), regarded as high-dimensional data, is capable of capturing the surface traits over an extensive wavelength range. HSI harbors the spectral traits associated with reflection or emission across diverse wavelengths [1–3]. Owing to its distinctive spectral attributes [4], it becomes possible for HSI to house more all-encompassing data [5–9]. Consequently, HSI finds extensive utilization in multiple domains. Specifically, it is applied in object detection as referred to in [10,11], classification [12–14], as well as feature learning [15]. However, HSIs are often contaminated by serious noise, which may originate from sensor defects, atmospheric interference, and external light reflections from the surface. Noise substantially degrades the imaging quality of HSI, thereby reducing its reliability in practical applications. Therefore, addressing the noise-induced degradation of HSI quality is a crucial step before applying HSI in real scenarios [16].

Up to now, scholars have put forward a large number of HSI denoising methods that are based on prior information or deep learning. Specific instructions are given as follows.

Prior-based techniques generally confine the solution space of this ill-posed issue by erecting specific assumptions or priors. These priors include total variation [17–19], sparsity [20], low-rank [21–24], and non-local similarity [25], among others. Such methods apply regularization constraints to the model for HSI denoising. Additionally, some approaches based on low-rank tensor representation leverage structural correlations and low-rank priors [26–28] to extract more inherent information from HSI. For example, Maggioni et al. [29] devised a block-matching 4-D (BM4D) model, capitalizing on the local correlations inside each HSI sub-cube. Similarly, Peng et al. [30] put forward a tensor dictionary learning model (TDL). Subsequently, Zhang et al. [20] devised an effective HSI recovery approach predicated on low-rank matrix. To combine spatial non-local similarity with global spectral low-rank characteristics, He et al. [25] introduced a unified spatial-spectral paradigm model (NGMeet) for HSI denoising. Xue et al. [31] built a non-local low-rank regularized tensor decomposition model. Overall, prior-based methods usually rely on the manually designed priors and require adjustments of hyperparameters depending on the

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specific scenarios. Such challenges impede their efficacy when applied to real-world datasets and necessitate considerable computational time along with resources [32].

In contrast, deep learning-based methods [33–37] have been verified to be more flexible and proficient at handling the HSI denoising problem by leveraging deep neural network models. In recent years, inspired by denoising methods for natural images, researchers have begun to apply deep learning techniques to HSI denoising. In contrast to conventional RGB and grayscale images, HSI harbors copious spectral details. However, this also gives rise to two prominent hurdles. Firstly, there is substantial redundancy within the spectral domain. Secondly, the task of capturing long-range contextual dependencies entails a hefty computational load. In recent studies, deep learning networks, particularly those based on convolutional neural networks (CNNs) and Transformer architectures [38], have demonstrated notable advantages in HSI denoising tasks.

For instance, Wei et al. [39] effectively explored the spatial-spectral dependencies and global spectral correlations. Pan et al. [40] progressively utilized the spectral and spatial components to extract the inherent spectral and spatial features of noise in HSI. Lai et al. [41] introduced a mixed attention mechanism that simultaneously considers spectral correlations and the interactions between spatial-spectral features at different scales. Additionally, Zhang et al. [42] devised a spatial-spectral gradient network (SSGN) for the purpose of removing the mixed noise present in HSI. Despite the CNN method having the potential for HSI restoration, existing works in this field still face several challenges. Firstly, the inherent weight-sharing strategy of CNN hampers the model's aptitude to spontaneously accommodate different inputs, which may affect performance when processing low-quality images. Secondly, constrained by the kernel size, the receptive field of individual convolutional layers is limited, which may substantially augment local bias. Despite the fact that certain studies have put forward alternative convolution designs, the CNN-based denoising approaches remain restricted by the size of the effective receptive field.

Compared to 2D images, the high-dimensional nature of HSI cubes poses a dilemma for deep learning. CNNs fail to capture long-range dependencies due to limited receptive fields, whereas Transformers, despite their global modeling capability, incur prohibitive quadratic complexity on massive HSI. This computational burden usually necessitates suboptimal window-based processing, which compromises the integrity of global spatial-spectral information. Therefore, the critical objective of this work is to achieve global spatial-spectral modeling with linear complexity, effectively resolving the trade-off between restoration quality and computational cost.

To address the limitations of the receptive field in CNN models, subsequent Transformer-based models [43–45] have made significant progress in modeling long-range dependencies and improving model performance. For instance, SERT [46] introduced a rectangular sliding window module and a low-rank module to perform HSI denoising in the local spatial-spectral context. HSDT [47] developed a mask-guided mechanism for HSI denoising. SST [48] employed a window attention module and global spectral attention to enhance denoising capabilities, aiming to capture spatial similarity and spectral dependencies. However, despite the ability of Transformers to capture global interactions, they face challenges in terms of computational and data efficiency, because they require large-scale datasets for training [49]. Therefore, effectively leveraging self-attention to explore the spatial-spectral non-local similarities in HSI remains a key challenge in current HSI denoising methods.

Many scholars combined data-driven and model-driven strategies for HSI denoising [50]. For instance, HIR-Diff [51] utilized the diffusion model to learn the distribution of HSI as the generation prior. It explicitly incorporates the smoothness prior constraint into the diffusion sampling process. Inspired by the physical property of spectral low-rankness, MST [52] used the self-attention along the spectral direction as a spectral correlation prior. SMDSNet [53] introduced a model-guided interpretable

network for HSI denoising. It considers the spatial redundancy, spectral low-rankness, and spectral-spatial correlations of HSIs. Nevertheless, due to the limitations of computational efficiency and memory usage, the development of such methods has not yet reached a mature stage.

To mitigate the efficiency bottleneck of Transformers while maintaining global modeling capabilities, the Retentive Network (RetNet) was recently introduced as a robust successor architecture. RetNet proposes a retention mechanism to replace standard self-attention, successfully achieving the 'impossible triangle' of parallel training, low-cost recurrent inference, and competitive performance. By incorporating an explicit distance-dependent decay prior, RetNet secures linear computational complexity ($O(N)$). RMT [54] is a representative work that successfully transfers RetNet to the field of vision; it introduced a general-purpose visual backbone network with a clear spatial prior. This unique characteristic makes it highly effective for processing long-sequence data, providing a theoretical foundation for efficient global modeling in high-dimensional HSI restoration. To address above issues, this work proposes the spatial-spectral global modeling retentive network (SSGMRN), which alternately aggregates spatial and spectral information to achieve more effective spatial-spectral feature representation. As shown in Fig. 1, in contrast to most of the advanced techniques, our methodology achieves superior results in terms of PSNR, runtime, and model parameter count.

Instead of standard self-attention, SSGMRN employs a dual-domain retention mechanism. In the spatial domain, a bidirectional decay mask based on Manhattan distance is constructed, which decomposes 2D spatial context into linear recurrent operations. In the spectral domain, a spectral decay mask is introduced to model inter-band correlations based on spectral proximity. This architecture fundamentally solves the trade-off between global modeling and efficiency. By explicitly introducing distance-dependent decay as an inductive bias, the model prioritizes spatially and spectrally neighboring information, while retaining the ability to capture long-range dependencies. Although RetNet was recently introduced into the hyperspectral domain for classification tasks, in the denoising task, how to utilize its linear complexity to handle the global restoration of high-dimensional spectra remains unexplored. SSGMRN is the first attempt to address this issue. The main contributions of this method are listed as follows:

- SSGMRN incorporates the spatial and spectral priors that decay with distance. It allows the model to more effectively retain spatial and spectral information, which varies with the distance, thereby improving HSI denoising performance.
- In the spatial domain, a two-dimensional bidirectional spatial decay matrix is designed based on the Manhattan distance between pixels. This decay matrix is combined with an attention

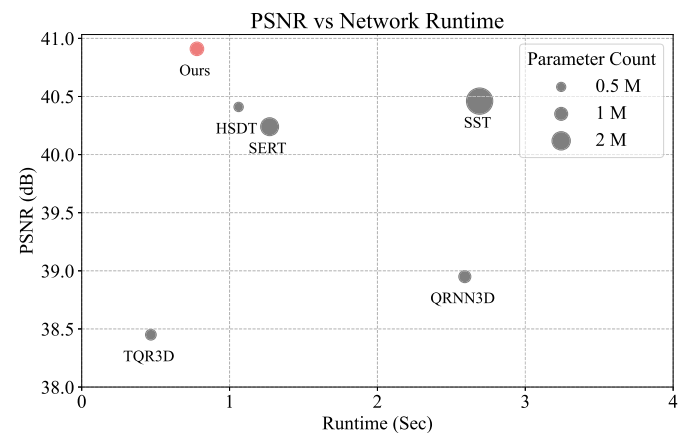


Fig. 1. Quantitative comparisons of PSNR, runtime, and model parameter counts between SSGMRN and other learning-based methods on the ICVL dataset under mixed noise.

decomposition mechanism, enabling efficient modeling of spatial information in HSI with reduced computational complexity.

- In the spectral domain, the proposed decay matrix effectively assists in modeling spectral information, enhancing the network's ability to learn spectral features across different bands in HSI.

2. Related work

2.1. Efficient transformers and linear attention

While standard Transformers have achieved remarkable success in global modeling, their quadratic computational complexity $O(N^2)$ poses a severe bottleneck for high-dimensional data like HSI. To mitigate this, various efficient variants have been proposed. Linear Transformers [55,56] utilize kernel-based feature maps to reduce complexity to $O(N)$. However, these linearization methods often disrupt the explicit modeling of relative positions, requiring additional, often complex, positional encoding strategies to retain local context. Other variants like Linformer [57] and Performer [58] employ low-rank approximations or random feature maps. Although efficient, they typically lack a mechanism to explicitly prioritize local neighborhoods within the global context, which is crucial for image restoration tasks where local textures and edges are as important as global consistency.

2.2. State space models (Mamba)

Recently, Structured State Space Models (SSMs), such as Mamba [59], have emerged as a powerful alternative for long-sequence modeling with linear complexity. Mamba excels at capturing continuous-time dynamics. However, applying 1D-based SSMs to 2D or 3D image data often requires specific scanning strategies, and their hidden state formulation is less intuitive for explicitly modeling the distance-decay property inherent in spectral correlations compared to attention-based models.

2.3. Retentive networks and rationale for SSGMRN

Retentive Networks (RetNet) [60] were proposed to bridge the gap between Transformers and RNNs, enabling parallel training, low-cost inference, and efficient long-term memory. RetNet is selected as the foundational architecture for HSI denoising primarily due to its

Explicit Decay Mechanism, which offers a unique advantage over Linear Transformers and Mamba for spectral data. HSI data exhibits strong spectral continuity, where the correlation between bands naturally decays as the spectral distance increases. RetNet incorporates a decay matrix directly into the retention formulation. This acts as an explicit, physically-interpretable prior that enforces the model to pay more attention to adjacent bands or pixels while still allowing long-range interactions. Unlike standard Linear Attention which treats all tokens as globally accessible with equal potential, RetNet's decay mechanism structurally enforces a locality bias within the global scope. This is critical for HSI denoising to avoid overfitting to distant, irrelevant spectral noise while preserving local spectral signatures. RetNet makes it feasible to process volumetric HSI data that is computationally prohibitive for standard Transformers.

3. Methodology

In this section, the noise degradation model pertaining to HSI is presented. Then, the overall structure of the proposed model is presented, along with the design of its individual modules. Finally, the training and optimization process of the model are described.

3.1. HSI noise degradation model

Let the noisy HSI be denoted as $\mathcal{Y} \in \mathbb{R}^{H \times W \times C}$, where H , W , and C represent the height, width, and spectral dimension, respectively. In general, the HSI noise degradation model could be approximated as:

$$\mathcal{Y} = \mathcal{X} + \mathcal{N} \quad (1)$$

where $\mathcal{X}$ stands for the noise-free HSI, $\mathcal{N}$ denotes the noise interference. In the real-world scenarios, HSI commonly suffers from diverse forms of noise contamination. Gaussian noise, stripe noise, deadline noise, impulse noise, as well as mixed noise are among the typical ones. The target of HSI denoising is to estimate the noise-free HSI from the noisy observations.

3.2. Model architecture and design

Fig. 2 presents the general framework of the proposed SSGMRN model for HSI denoising. The model consists of a feature extractor, a

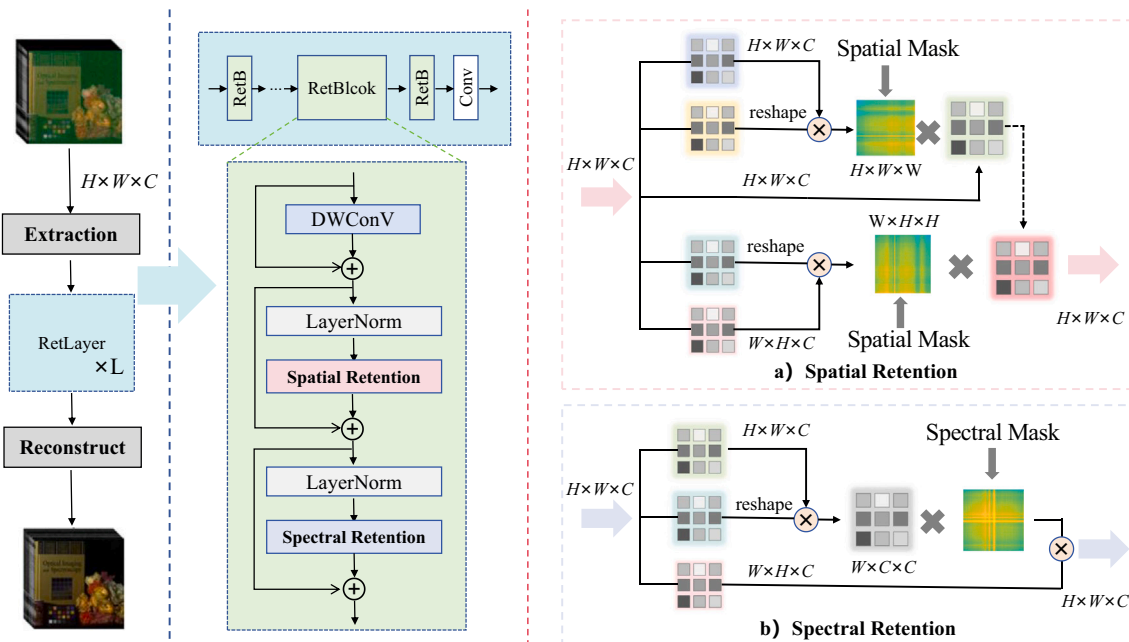


Fig. 2. Overall structure of the proposed SSGMRN model.

reconstructor, and three residual retention layers (RRLs) connected by skip connections. The detailed descriptions of the design of each module are provided below.

3.2.1. Retention network

In recent years, along with the swift advancement of large-scale language modeling, a new network architecture known as the retention network (RetNet) has emerged. RetNet leverages a distance-dependent temporal decay matrix to provide an explicit temporal prior for text data. It employs multi-scale gated retention mechanisms to preserve contextual information, effectively guiding long-range dependencies without compromising computational efficiency. By transforming the observed unidirectional and one-dimensional temporal decay into bidirectional and two-dimensional spatial decay, RetNet introduces a clear spatial prior based on Manhattan distance. It excels in processing temporal information and demonstrates the strong global modeling capability for non-linear features.

The RetNet introduces a retention mechanism for sequence modeling, incorporating temporal decay into language models, which is absent in Transformer models. The RetNet could be expressed as:

$$Q = (XW_Q) \odot \Theta, \quad K = (XW_K) \odot \bar{\Theta}, \quad V = XW_V \quad (2)$$

$$\Theta(n) = e^{in\theta} \quad D_{nm} = \begin{cases} \gamma^{n-m}, & n \geq m \\ 0, & n < m \end{cases} \quad (3)$$

$$\text{Retention}(X) = (QK^T \odot D)V \quad (4)$$

where Θ represents the conjugate transpose of $\bar{\Theta}$. $W_Q, W_K, W_V \in \mathbb{R}^{[x] \times [x]}$ and $D \in \mathbb{R}^{[x] \times [x]}$ incorporate the causal masking and exponential decay. This formulation captures the relative distances within a one-dimensional sequence and introduces an explicit temporal prior in text data.

By establishing a clear spatial prior related to Manhattan distance, the observed unidirectional and one-dimensional temporal decay is transformed into bidirectional and two-dimensional spatial decay. In RetNet, due to the causal nature of textual data [54], retention is unidirectional, meaning that each token could only attend to preceding tokens, not succeeding ones. This characteristic is not applicable to tasks that lack a causal structure, such as HSI denoising. Therefore, the bidirectional form of the RetNet could be represented as:

$$D_{nm}^{Bi} = \gamma^{|n-m|} \quad (5)$$

In a 2D image, spatial information is crucial. Within the context of the image, each pixel is uniquely positioned in the plane using two-dimensional coordinates. For the n -th label, it is represented by (x_n, y_n) . To accommodate it, each element in matrix D is adjusted to represent the Manhattan distance based on the respective pair of label coordinates. The updated definition of matrix is fixed as follows:

$$D_{nm}^{2d} = \gamma^{|x_n - x_m| + |y_n - y_m|} \quad (6)$$

It is worth noting that the proposed SSGMRN stands out due to its advanced temporal decay feature, achieving a balance between linear computational complexity and effective modeling of long-term dynamics. SSGMRN enhances the original Transformer-based model by introducing a spatial spectral retention block, making it more suitable for efficiently and effectively denoising multi-dimensional HSI.

SSGMRN uniquely incorporates a distance-decaying mask as both spatial and spectral priors, enabling the model to more effectively retain the spatial and spectral information as it varies with distance. In our decay mask, the attention score decreases to a greater extent for labels that are farther away from the target label. This property allows the target token to capture global information while assigning different levels of global attention to tokens at varying distances. By considering both spectral correlations and non-local spatial similarities, SSGMRN effectively exploits the rich spectral features of HSI while maintaining the

spatial structural integrity and spectral fidelity of HSI. The following sections provide the detailed explanations of the proposed SSGMRN.

3.2.2. Spatial retention module

How to effectively utilize the spatial information is one of the key factors in improving HSI denoising performance. To address this issue, SSGMRN introduces an innovative spatial retention module. It aims to leverage the explicit distance priors to adapt to the two-dimensional spatial decay characteristics in HSI, while also reducing the computational cost. The core idea of the spatial retention module is to enhance denoising performance by considering the spatial relationships between each pixel in HSI. This module first computes the attention scores in both the horizontal and vertical directions within the HSI. Specifically, the module calculates the spatial attention scores for each pixel, taking both horizontal and vertical spatial information into account. SSGMRN enables the network to more accurately preserve the spatial structure of HSI during denoising, rather than treating all pixels independently. In particular, the attention matrices in the horizontal and vertical directions are defined as follows:

$$\text{Attn}_H = \text{Softmax}(Q_H K_H^T) \odot D^{Hnm} \quad (7)$$

$$\text{Attn}_W = \text{Softmax}(Q_W K_W^T) \odot D^{Wnm} \quad (8)$$

The one-dimensional decay matrices are defined as $D_{Hnm} = \gamma^{|y_n - y_m|}$ and $D_{Wnm} = \gamma^{|x_n - x_m|}$, where γ is the decay factor. $|y_n - y_m|$ and $|x_n - x_m|$ represent the distances for pixel (n, m) along the vertical and horizontal directions, respectively. In this manner, the decay matrices effectively capture the spatial distribution characteristics of HSI, allowing the network to account for the spatial relationships between pixels located at different positions when processing HSI.

Next, the spatial retention module applies the one-dimensional bidirectional decay matrices to the attention weights, enabling decay operations in the spatial domain:

$$\text{SpatialRetention}(X) = \text{Attn}_H (\text{Attn}_W V)^T \quad (9)$$

As shown in Fig. 3, spatial attention maps, Attn_W and Attn_H , are generated along the height (H) and width (W) directions of HSI, respectively. Guided by the spatial decay mask, the generated horizontal and vertical attention maps are then fused together. This decomposition method effectively preserves explicit spatial prior information. In the spatial retention module, we propose a spatial decay-based self-attention mechanism, in which the feature maps gradually diminish as the spatial distance increases. The spatial decay mechanism focuses on neighboring pixels with reference information and gathers richer contextual information from non-local regions. Furthermore, prior to the spatial retention module, we introduce a local context enhancement module that leverages depth-wise separable convolutions (DWConv), to further enhance the feature representation capability.

3.2.3. Spectral preservation module

Compared to natural images, HSI contains more extensive spectral information. Therefore, how to effectively utilize the spectral information is crucial for guiding HSI recovery. In HSI, SSGMRN also introduces a decay mechanism along the spectral dimension. This means that the model not only focuses on spatially adjacent pixels, but also on adjacent spectral bands, thereby enhancing the interaction and integration between spatial and spectral features. For HSI, the spatial retention module could adaptively adjust the attention weights based on the spatial and spectral relationships at different positions. For specific spectral bands, the model dynamically adjusts its attention weights based on the spectral correlation of adjacent bands. This weighting strategy allows for a more precise fusion of spatial and spectral features, particularly when dealing with HSIs that have the high spectral redundancy, effectively preventing excessive attenuation or loss of information. Specifically, the spectral attention map could be guided using a spectral decay mask matrix. It is

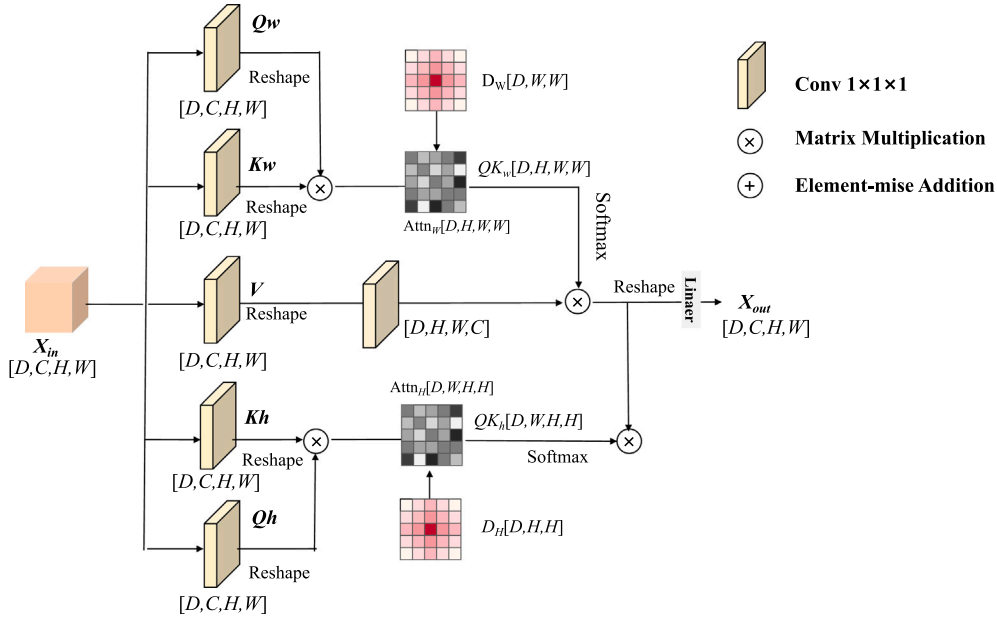


Fig. 3. Structure of the spatial retention module in SSGMRN.

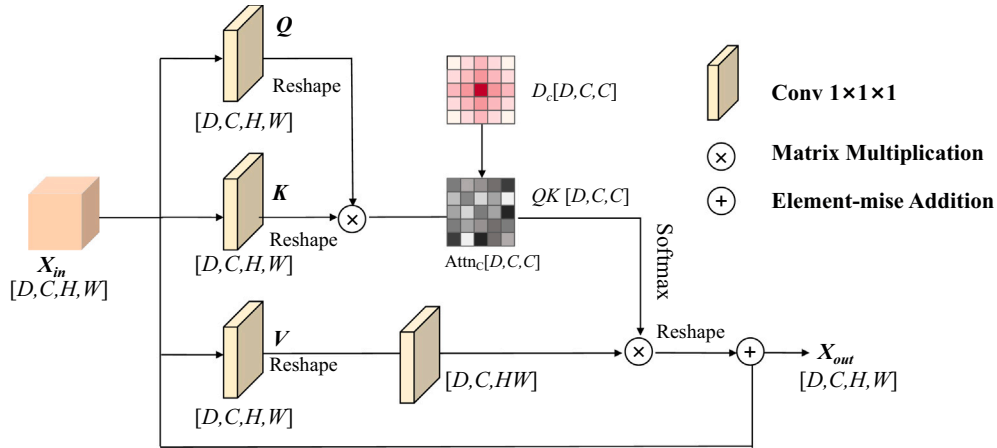


Fig. 4. Structure of the spectral retention module in SSGMRN.

represented as Eq. (10), where the decay is based on the band interval distance:

$$\text{SpectralRetention}(X) = (\text{Softmax}(Q_C K_C^T) \odot D^C) V \quad (10)$$

In the proposed SSGMRN, the spectral retention module efficiently models long-range spectral dependencies with linear complexity, by fully exploiting global spectral correlations and neighboring spectral information. This module effectively captures the deep relationships between different bands in HSI, thereby enhancing the models performance in spectral information extraction. The specific structure of the spectral retention module is displayed in Fig. 4.

As shown in Fig. 4, the spectral decay mask is critical for guiding the generation of the spectral attention map. By employing this mask, the module adaptively adjusts the attention weights of individual bands, allowing the network to focus on spectral regions most relevant to the target task. Specifically, the mask incorporates long-range spectral dependencies, enabling the model to capture global spectral correlations while diminishing the disturbances caused by redundant spectral traits. Furthermore, this design maintains an efficient computational

framework with linear complexity, ensuring robust spectral feature extraction even under complex spectral redundancy.

3.3. Model training and optimization

3.3.1. Loss function

When it comes to HSI denoising, the loss function is generally specified with the aid of the L1 norm or L2 norm. Aiming to attain a balance between removing noise and maintaining details, this work adopts the L2 norm as the loss function for the proposed model, which is defined as follows:

$$Loss = \frac{1}{N} \sum_{i=1}^N \|X_i - X_i^{gt}\|_2 \quad (11)$$

where X_i represents the restored HSI. X_i^{gt} denotes the noise-free HSI.

3.3.2. Training and optimization

SSGMRN uses the ICVL dataset [61] for model training and optimization. This dataset contains 201 HSIs, each with spatial dimensions of 1392×1300 pixels, covering 31 bands ranging between 400 nm and

700 nm. During the training process, 100 HSIs are employed for training and 40 HSIs for testing. To enhance training performance, various data augmentation techniques, such as random cropping, rotation, and flipping are applied. This results in images being resized to 64×64 , generating approximately 53,000 training samples. For the testing process, the images are centrally cropped to a spatial size of 512×512 .

The training consists of 50 epochs, with the first 20 epochs focusing on training with Gaussian noise, including both fixed-level and random Gaussian noise. The remaining 30 epochs involve training with different combinations of complex noise scenarios. Since real HSIs are usually contaminated by various complex noises, such as Gaussian noise, sparse noise, or their combinations, five different complex noise cases are inserted into the pure HSIs to emulate these practical noise conditions in reality. The specific noise simulation cases are outlined as follows:

Case 1: Non-iid Gaussian Noise: Gaussian noise with zero mean is added to each band, and the noise intensity is randomly set within the range of [10, 70].

Case 2: Gaussian + Stripe Noise-Based on Case 1: Gaussian noise is applied to all the bands. Additionally, 10 random bands are contaminated with stripe noise, and the proportion of simulated stripe noise in these bands is set between [5%, 15%].

Case 3: Gaussian + Deadline Noise-Similar to Case 2: while instead of stripe noise, deadline noise is applied to the 10 randomly selected bands.

Case 4: Gaussian + Impulse Noise-Based on Case 1: Gaussian noise is applied to all the bands. Additionally, 10 random bands are contaminated with impulse noise, with the intensity of the impulse noise ranging from [10%, 70%].

Case 5: Mixed Noise-Based on Case 1: Gaussian noise is applied to all the bands. Furthermore, 10 random bands are contaminated with one of the noise types from Cases 1 to 4.

The proposed model is tuned with the Adam optimizer [62]. The learning rate is held constant at 1×10^{-4} , and the batch size is configured as 16. From a software perspective, the proposed model is realized through the PyTorch deep learning framework.

On the hardware side, the model runs on an NVIDIA RTX 4090 GPU, an Intel i9-12900K CPU, and 32 GB of RAM.

4. Experiments

In this section, we compare the results of the state-of-the-art methods and the proposed method in both simulated and real HSI denoising experiments. The details are described as follows:

4.1. Experimental setup

4.1.1. Test datasets

The test datasets consist of two parts: simulated experiments and real experiments. We conducted simulated experiments on the ICVL dataset, which includes Gaussian denoising of HSI at different noise levels $\sigma = 30, \sigma = 50, \sigma = 70$ and blind noise) as well as complex noise

denoising. The simulated experiments allow for the evaluation of various methods under different noise intensities and types, helping to assess the robustness and performance of diverse noise scenarios. In addition, actual denoising experiments are implemented on HSI data from Urban ($307 \times 307 \times 188$), GF-5 ($300 \times 300 \times 155$), and Ziyuan-1 02D ($200 \times 200 \times 90$) datasets. Some of the experimental data have been trimmed.

4.1.2. Comparison methods and evaluation metrics

The comparison methods include SOTA approaches for HSI denoising, encompassing both traditional prior-based methods, such as LRTV [42], NGMeet [25], and STCR [63], as well as deep learning-based methods: QRNN3D [39], TRQ3D [64], SERT [46], SST [65], and HSDT [47]. Notably, TRQ3D, SERT, SST, and HSDT are based on Transformer architectures, which are included to highlight the denoising advantages of the proposed method. For the fair comparison, the prior-based methods are fine-tuned with the default settings provided via the parameter guidelines from the respective papers. The deep learning-based methods are retrained to ensure the optimal HSI denoising performance.

We quantitatively evaluate the HSI denoising performance of all the methods in the simulated experiments, using metrics such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Spectral Angle Mapper (SAM). Higher PSNR and SSIM values, along with lower SAM values, indicate the better denoising performance.

4.2. Simulated experiments

4.2.1. Gaussian noise removal on the ICVL dataset

In this experiment, we apply Gaussian noise of varying intensities (including 30, 50, 70, and random intensities ranging from 30 to 70) to the ICVL dataset, to evaluate the performance of different HSI denoising methods. The quantitative results are summarized in Table 1.

The denoising comparison results on the ICVL dataset at different levels of Gaussian noise intensity are presented in Table 1. Among the HSI denoising methods based on prior-based models, although STCR and NGMeet perform well at low noise levels, their performance deteriorates significantly as the noise intensity increases. This is because these methods primarily rely on spectral prior models or local features, which limits their ability to capture global spectral correlations and long-range dependencies. As the noise intensity increases, their performance is heavily influenced by noise, leading to the loss of image details and spectral distortions.

As listed in Table 1, deep learning-based methods achieve superior performance across various noise levels. In addition, to provide a comprehensive visual assessment, Fig. 5 presents the denoising results of different methods. At a high noise level of $\sigma = 70$, SST exhibits spectral distortion, primarily due to its limitations in noise suppression, making it difficult to handle complex noise patterns. At higher noise levels, the proposed SSGMRN model, with its spectral retentive module, adapts by introducing a spectral decay mask. This allows the proposed model to adjust the importance of each band, and accurately capture long-range

Table 1
Quantitative denoising results on the ICVL dataset for different methods under different Gaussian noise intensities.

Method	30			50			70			blind		
	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM
Noisy	18.59	0.1110	0.6840	14.15	0.0484	0.8685	11.22	0.0267	0.9941	16.54	0.1039	0.7687
RCTV	36.76	0.9325	0.1239	36.91	0.8915	0.2173	32.23	0.8863	0.1514	34.18	0.8238	0.2201
NGMeet	38.21	0.9451	0.1310	37.81	0.9172	0.1946	34.41	0.9066	0.1201	34.88	0.8671	0.2075
STCR	37.81	0.9365	0.1313	35.43	0.7128	0.4652	31.03	0.7687	0.2492	34.91	0.8769	0.2727
QRNN3D	39.94	0.9498	0.0941	38.28	0.9309	0.0968	36.32	0.8920	0.1280	38.95	0.9356	0.1045
TRQ3D	40.55	0.9607	0.0550	38.45	0.9403	0.0643	36.87	0.9188	0.0883	39.49	0.9491	0.0720
SERT	42.41	0.9703	0.0501	40.24	0.9540	0.0598	37.81	0.9317	0.0825	41.21	0.9602	0.0659
SST	42.48	0.9708	0.0483	40.46	0.9559	0.0543	36.83	0.9020	0.1433	41.00	0.9508	0.0972
HSDT	42.46	0.9692	0.0484	40.41	0.9537	0.0553	38.78	0.9372	0.0721	41.41	0.9605	0.0603
Ours	42.51	0.9660	0.0566	40.91	0.9562	0.0611	39.45	0.9440	0.0677	41.83	0.9611	0.0609

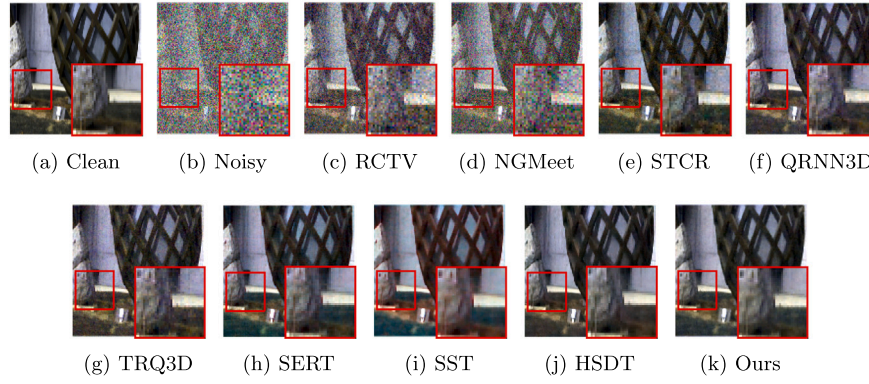


Fig. 5. Comparison of visual denoising results on the ICVL dataset (HSI: Bulb_0822 - 0909, bands: 30, 25 and 10) under Gaussian noise with $\sigma = 70$.

Table 2

The quantitative denoising performance of different methods on the ICVL dataset under five mixed noise scenarios.

Method	Noniid + G			Stripe + G			Deadline + G			Impulse + G			Mixture + G		
	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM
Noisy	17.79	0.1616	0.7915	17.75	0.1598	0.7838	17.68	0.1624	0.7942	15.11	0.1229	0.8415	14.24	0.1022	0.8549
RCTV	33.14	0.8895	0.0753	33.99	0.8482	0.1912	33.21	0.8007	0.1333	36.59	0.8960	0.1884	26.11	0.7822	0.1546
NGMeet	35.33	0.8880	0.0697	37.04	0.9180	0.1310	34.41	0.8530	0.1058	35.26	0.8599	0.2067	31.16	0.8856	0.1344
STCR	34.45	0.9090	0.1046	32.93	0.7262	0.2998	33.78	0.8823	0.2961	32.70	0.7493	0.2088	25.76	0.7884	0.1806
QRNN3D	40.56	0.9612	0.0732	38.88	0.9358	0.0910	40.28	0.9602	0.0745	40.39	0.9603	0.0740	37.87	0.9301	0.0920
TRQ3D	41.62	0.9721	0.0445	41.56	0.9718	0.0453	41.41	0.9708	0.0456	39.98	0.9552	0.0653	39.19	0.9527	0.0728
SERT	43.23	0.9767	0.0409	40.27	0.9535	0.0709	42.82	0.9763	0.0428	43.00	0.9764	0.0422	38.60	0.9472	0.0857
SST	41.90	0.9694	0.0418	41.64	0.9689	0.0432	40.92	0.9683	0.0470	38.91	0.9482	0.0686	36.80	0.9326	0.0887
HSDT	42.89	0.9742	0.0399	42.60	0.9736	0.0424	42.40	0.9732	0.0421	40.02	0.9528	0.0815	38.89	0.9479	0.0675
Ours	43.40	0.9792	0.0377	43.15	0.9785	0.0400	42.88	0.9780	0.0397	40.92	0.9639	0.0605	39.25	0.9537	0.0633

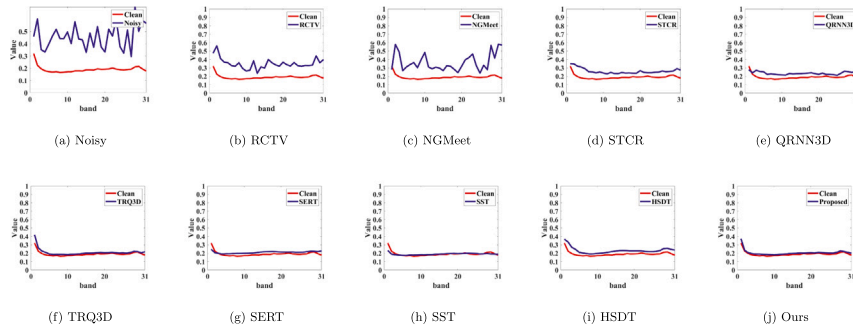


Fig. 6. Comparison of spectral curves at (150,75) pixel of ICVL dataset (HSI: Bulb_0822 - 0909) under Gaussian noise with $\sigma = 70$.

spectral dependencies. Compared to other methods, SSGMRN better preserves the global spectral structure of the image during the denoising process, avoiding the issue of traditional methods neglecting crucial information at low signal-to-noise ratios.

4.2.2. Denoising results on the ICVL dataset under mixed noise conditions

In the mixed noise removal experiments, to evaluate the denoising performance under different noise scenarios, various types of noise are introduced into the ICVL dataset to simulate multiple mixed noise environments. A total of five cases are designed, with the detailed descriptions provided in Table 2.

Comparing the performance of different methods, Table 2 presents the quantitative denoising results for the ICVL dataset under five mixed noise scenarios. In most scenarios, the proposed SSGMRN outperforms other methods in terms of mixed noise removal. Additionally, the visual comparison of denoising results under mixed noise conditions on the ICVL dataset is displayed in Fig. 7. Traditional methods exhibit a

significant decline in denoising performance when dealing with complex noise, primarily due to their reliance on prior-based optimization strategies. In contrast, the proposed method excels at preserving spectral consistency through band modeling, enabling more effective handling of complex noise environments.

4.2.3. Spectral consistency analysis

To comprehensively evaluate the spectral fidelity of the restored images, the spectral curves of the denoised results against the ground truth were compared. Fig. 6 illustrates the spectral curves of a representative pixel at position (150, 75) from the ICVL dataset under heavy Gaussian noise ($\sigma = 70$). The results obtained by the RCTV and MGMeet methods contain a considerable amount of noise residue. Several deep learning methods could obtain relatively smooth spectral curves, which are quite similar to those of clean images. Especially, TRQ3D, SERT, SST and the proposed SSGMRN are the most smoothing and closely match the original image. Among them, the smoothing effects of TRQ3D, SERT, and

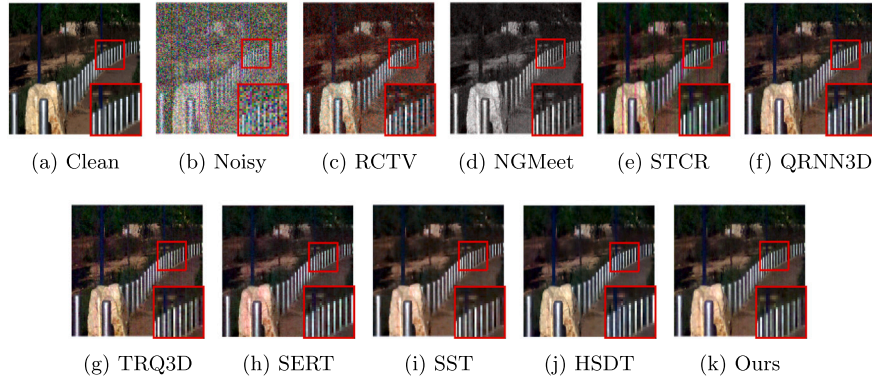


Fig. 7. Comparison of visual denoising results on the ICVL dataset (HSI: Lehavim_0910 - 1708, bands: 22, 7 and 4) under mixed noise scenarios.

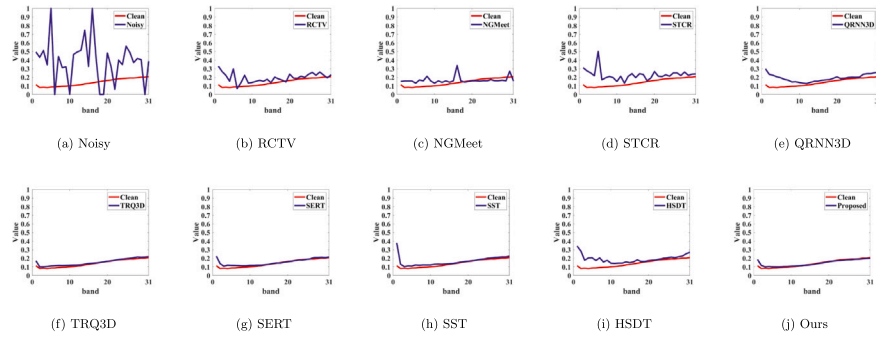


Fig. 8. Comparison of spectral curves at (150,75) pixel of ICVL dataset (HSI: Lehavim_0910 - 1708) under mixed noise scenarios.

SST in the first few bands all showed either excessive or insufficient results. The proposed SSGMRN had the curve that was most similar to the clean spectral curve. Furthermore, Fig. 8 presents the spectral curve comparison under the more challenging mixed noise scenario. The original spectral information is severely corrupted by complex noise patterns, leading to sharp spikes and intensity distortions. In Fig. 8, SSGMRN demonstrates superior robustness, compared to the other state-of-the-art methods. The performance of TRQ3D and the proposed SSGMRN is comparable in the first few bands. However, in the final few bands, the proposed SSGMRN is more consistent with the original curve. This strong spectral preservation capability is attributed to the spectral retention module, which effectively captures global spectral correlations and utilizes clean bands to guide the restoration of corrupted bands.

4.3. Real-world experiments

To validate the robustness of the proposed method under real-world noise conditions, this section conducts real HSI denoising experiments using the Urban, GF-5, and Ziyuan-1 02D HSI datasets. For deep learning-based approaches, the ICVL pre-trained model was used on the three real datasets. For testing purposes, all the real HSIs are sliced into 31 bands for processing.

4.3.1. Urban HSI data

The spatial size of the Urban HSI data is 307×307 , with 188 bands, and a spectral range from 400 nm to 2500 nm. In this HSI, several bands are affected by atmospheric radiation and other interferences, leading to severe mixed noise contamination. The noise types include deadline, stripe, and Gaussian noise. Consequently, removing the mixed noise in this HSI is a significant challenge. Fig. 9 shows the real experimental results of the comparison algorithms and the proposed methods on the Urban HSI data. Traditional prior-based methods exhibit varying degrees of noise residuals, especially in the removal of striping

noise. In contrast, deep learning-based methods, with their powerful feature-learning capability, achieve better HSI denoising performance.

4.3.2. GF-5 HSI data

The GF-5 HSI data are acquired from the Gaofen-5 satellite. The size of the dataset is $300 \times 300 \times 155$. In addition to being contaminated by random noise, this GF-5 HSI data also suffers from significant coarse striping noise. The pseudo-color denoising results are shown in Fig. 10. The QRNN3D method produces over-smoothing results. In contrast, the proposed method, based on an attention mechanism, achieves the best performance in both noise removal and texture detail recovery. This further validates the effectiveness of the proposed method.

4.3.3. Ziyuan-1 02D HSI data

We use Ziyuan-1 02D HSI with a spatial size of 200×200 pixels and 90 bands as experimental data. This dataset is severely contaminated by mixed noise, resulting in a significant degradation of image quality. The pseudo-color denoising results of the Ziyuan-1 02D HSI data are portrayed in Fig. 11. Compared to other SOTA algorithms, the proposed method demonstrates a clear advantage in mixed noise removal and texture detail recovery, due to its spatial and spectral retentive mechanisms. In summary, the proposed method produces superior denoising results across all the three real HSI datasets, further validating its robustness and reliability in HSI denoising.

5. Discussion

5.1. Ablation study

To gauge the efficacy of the modules in the proposed technique, we execute the ablation study by systematically removing or adding each module and assessing its impact on denoising performance. All the models are trained in the same manner and evaluated on a non-iid dataset. The results are presented in Table 3. The introduction of the spectral

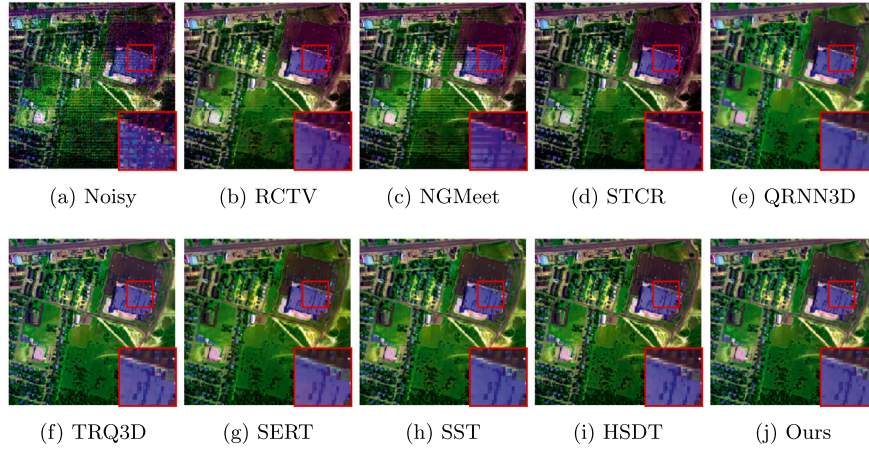


Fig. 9. Real denoising results on bands (188,104,1) of the Urban HSI data.

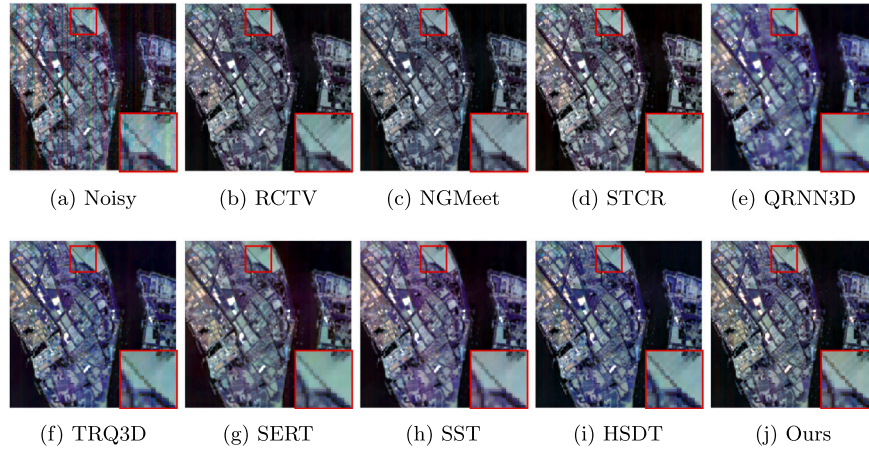


Fig. 10. Real denoising results on bands (152, 96, 43) of the GF-5 HSI data.

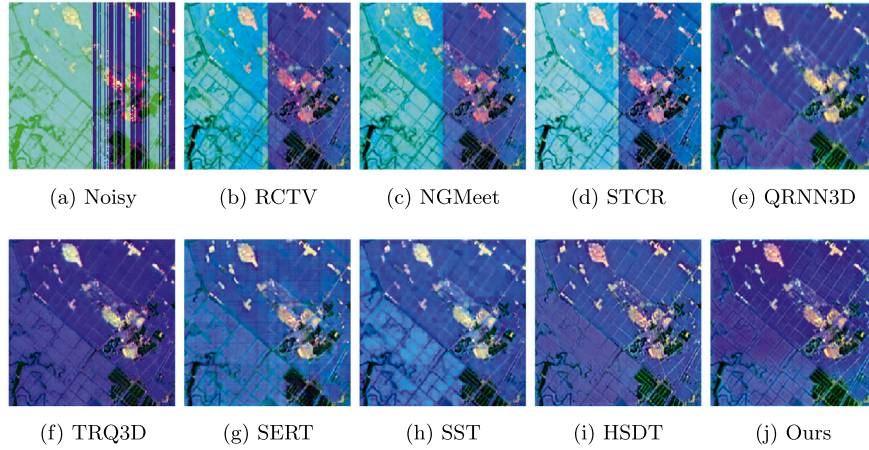


Fig. 11. Real denoising results on bands (80, 56,1) of the Ziyuan-1 02D HSI data.

retentive module and the spatial retentive module both improves the denoising performance. Furthermore, replacing the feedforward network (FFN) module with the spectral retentive module significantly enhances the model's ability to learn spectral features while simultaneously reducing the number of model parameters. Additionally, the DWConv module effectively improves the model's ability to learn local features while using fewer parameters.

5.2. Model complexity analysis

In Table 4, we compare the performance, parameter count, and average runtime of different deep learning-based methods on the ICVL dataset. By incorporating an attention mechanism in the spectral dimension, the proposed model achieves the optimal HSI denoising performance with fewer parameters. This advantage stems from its ability to

Table 3

Ablation study on the effectiveness of components within the proposed denoising method (ICVL dataset, non-iid noise).

LSC	Params (M)	PSNR (dB)	SSIM	SAM
Baseline + W/O spectral	0.68	41.80	0.9713	0.0474
Baseline + W/O spatial	0.40	40.74	0.9647	0.0745
Baseline + W/O DWConv	1.06	42.21	0.9723	0.0475
Baseline + W ffn + W/O spectral	1.67	42.76	0.9748	0.0373
baseline	1.08	43.40	0.9792	0.0377

Table 4

Comparison of model complexity via different deep learning methods (ICVL dataset, Gaussian noise level = 50).

Metric	QRNN3D	TQR3D	SERT	SST	HSDT	Ours
PSNR (dB)	38.95	38.45	40.24	40.46	40.41	40.91
Params (M)	0.86	0.68	1.91	4.10	0.52	1.08
Runtime (s)	2.5910	0.4678	1.2709	2.6911	1.0613	0.7896

Table 5

Analysis of different embedding dimensions on model's denoising performance.

Dim	Params (M)	PSNR	SSIM	SAM	Time(s)
48	0.63	37.06	0.9300	0.0773	0.6139
64	1.08	38.67	0.9450	0.0641	0.7078
96	2.34	37.99	0.9403	0.0692	0.9107

effectively reduce computational complexity, thereby enhancing both the efficiency and performance of the model. As shown in Table 4, the proposed method achieves the best denoising performance while incurring the minimal runtime.

Although TQR3D exhibits the lowest computational cost, its restoration quality lags significantly behind state-of-the-art performance. Conversely, while the methods based on transformers achieve competitive PSNR, they suffer from heavy parameter counts and slow inference due to the quadratic complexity of self-attention. Scaling these transformers to match SSGMRN's efficiency is challenging. SSGMRN achieves a Pareto-optimal balance, delivering the highest PSNR among compared methods, while maintaining a runtime and parameter count significantly lower than other high-performance counterparts, attributed to the linear complexity of the retentive mechanism.

5.3. Embedding dimension analysis

The impact of different embedding dimensions on the models feature extraction capability is significant. To address this problem, we analyze the effect of various embedding dimensions on model's denoising performance, as listed in Table 5. By setting the embedding dimension to 64, our model achieves the optimal performance. This choice enhances the model's performance without excessively increasing computational complexity, thus striking a balance between computational efficiency and effectiveness.

6. Conclusion

This paper proposes the SSGMRN model, which incorporates spectral and spatial retentive modules for HSI denoising. The model integrates explicit spatial and spectral attenuation mechanisms, thereby enhancing the ability to capture global spatial and spectral features. Additionally, the guidance provided by the attenuation matrices directs the network to focus on nearby spatial and spectral information. The explicit spatial attention mechanism, based on the spatial attenuation matrix and attention decomposition, operates with linear complexity. Through the incorporation of explicit spatial prior attention and spectral attention

mechanisms, SSGMRN strengthens the spatial-spectral interactions, significantly improving its perceptual capability. Both simulated and real experiments demonstrate that, compared to SOTA methods, SSGMRN delivers superior performance in HSI denoising. However, the reliance on fixed decay hyperparameters may limit the model's adaptability to diverse spectral correlations. Besides, the computational overhead remains a consideration for extremely large-scale HSI data. Future work will focus on introducing learnable decay mechanisms and optimizing the architecture.

CRedit authorship contribution statement

Qiang Zhang: Writing – original draft, Validation, Methodology. **Wenjing Gao:** Writing – original draft, Software, Methodology. **Weizhen Sun:** Writing – review & editing, Resources. **Yushuai Dong:** Writing – original draft, Visualization, Methodology. **Yaming Zheng:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] Y. Li, W. Chu, Y. Liu, D. Ma, Hyperspectral imaging through scattering media via physics-informed learning, *Opt. Laser Technol.* 170 (2024) 110299.
- [2] X. Cao, X. Wang, X. Dun, Y. Lian, X. Cheng, X. Hao, Cross-domain-aware deep unfolding transformer for hyperspectral image super-resolution, *Pattern Recognit.* 172 (2026) 112374.
- [3] X. Cao, Y. Lian, J. Li, K. Wang, C. Ma, Unsupervised multi-level spatio-spectral fusion transformer for hyperspectral image super-resolution, *Opt. Laser Technol.* 176 (2024) 111032.
- [4] Q. Wang, Q. Zhang, H. Xie, Z. Liu, Y. Dong, Sgd-sst: seamless global daily sea surface temperature products reconstruction and validation via deep spatio-temporal fusion model, *Expert Syst. Appl.* 293 (2025) 128703.
- [5] J. Tranon, R. d'Andrimont, A. Maignard, P. Defourny, Survey of current hyperspectral earth observation applications from space and synergies with sentinel-2, in: 2017 9th International Workshop on the Analysis of Multitemporal Remote Sensing Images (MultiTemp), 2017, pp. 1–8.
- [6] M.S. Mohd Asaari, P. Mishra, S. Mertens, S. Dhondt, D. Inzé, N. Wuyts, P. Scheunders, Close-range hyperspectral image analysis for the early detection of stress responses in individual plants in a high-throughput phenotyping platform, *ISPRS J. Photogramm. Remote Sens.* 138 (2018) 121–138.
- [7] C. Weber, R. Aguejda, X. Briottet, J. Avala, S. Fabre, J. Demuyne, E. Zenou, Y. Deville, M.S. Karoui, F.Z. Benhalouche, S. Gadal, W. Ourghemmi, C. Mallet, A.L. Bris, N. Chehata, Hyperspectral imagery for environmental urban planning, in: IGARSS 2018 - 2018 IEEE International Geoscience and Remote Sensing Symposium, 2018, pp. 1628–1631.
- [8] M. Shimoni, R. Haelterman, C. Perneel, Hyperspectral imaging for military and security applications: combining myriad processing and sensing techniques, *IEEE Geosci. Remote Sens. Mag.* 7 (2) (2019) 101–117.
- [9] Q. Jin, Y. Ma, X. Mei, X. Dai, H. Li, F. Fan, J. Huang, Gaussian mixture model for hyperspectral unmixing with low-rank representation, in: IGARSS 2019 - 2019 IEEE International Geoscience and Remote Sensing Symposium, 2019, pp. 294–297.
- [10] X. Li, M. Ding, A. Pizurica, Fully group convolutional neural networks for robust spectral-spatial feature learning, *IEEE Trans. Geosci. Remote Sens.* 60 (2022) 1–14.
- [11] R. Dian, T. Shan, W. He, H. Liu, Spectral super-resolution via model-guided cross-fusion network, *IEEE Trans. Neural Netw. Learn. Syst.* 35 (2023) 10059–10070.

- [12] X. Zhao, R. Tao, W. Li, W. Philips, W. Liao, Fractional gabor convolutional network for multisource remote sensing data classification, *IEEE Trans. Geosci. Remote Sens.* 60 (2022) 1–18.
- [13] R. Dong, Z. Wang, W. Wang, R. Zheng, High-fidelity reconstruction of compressed hyperspectral images based on physical mechanism guided selective attention network, *Opt. Laser Technol.* 191 (2025) 113268.
- [14] P. Zhu, J. Liu, Channel-wise transformer with spectral-spatial gated self-attention for hyperspectral image classification, *Opt. Laser Technol.* 193 (2026) 114182.
- [15] C. Wu, B. Du, L. Zhang, Hyperspectral anomalous change detection based on joint sparse representation, *ISPRS J. Photogramm. Remote Sens.* 146 (2018) 137–150.
- [16] Q. Zhang, J. Zhu, Y. Huang, Q. Yuan, L. Zhang, Beyond being wise after the event: combining spatial, temporal and spectral information for himawari-8 early-stage wildfire detection, *Int. J. Appl. Earth Obs. Geoinf.* 124 (2023) 103506.
- [17] W. He, H. Zhang, H. Shen, L. Zhang, Hyperspectral image denoising using local low-rank matrix recovery and global spatial-spectral total variation, *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 11 (3) (2018) 713–729.
- [18] W. He, H. Zhang, L. Zhang, H. Shen, Total-variation-regularized low-rank matrix factorization for hyperspectral image restoration, *IEEE Trans. Geosci. Remote Sens.* 54 (1) (2016) 178–188.
- [19] Q. Zhang, Y. Zheng, Y. Dong, C. Yu, Q. Yuan, Hyperspectral image mixed noised removal via jointly spatial and spectral difference constraint with low-rank tensor factorization, *Eng. Appl. Artif. Intell.* 149 (2025) 110508.
- [20] H. Zhang, W. He, L. Zhang, H. Shen, Q. Yuan, Hyperspectral image restoration using low-rank matrix recovery, *IEEE Trans. Geosci. Remote Sens.* 52 (8) (2014) 4729–4743.
- [21] Y. Chang, L. Yan, S. Zhong, Hyper-laplacian regularized unidirectional low-rank tensor recovery for multispectral image denoising, in: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017, pp. 5901–5909.
- [22] Q. Zhang, Y. Dong, Q. Yuan, M. Song, H. Yu, Combined deep priors with low-rank tensor factorization for hyperspectral image restoration, *IEEE Geosci. Remote Sens. Lett.* 20 (2023) 1–5.
- [23] F. Fan, Y. Ma, C. Li, X. Mei, J. Huang, J. Ma, Hyperspectral image denoising with superpixel segmentation and low-rank representation, *Inf. Sci.* 397–398 (2017) 48–68.
- [24] X. Fu, Y. Guo, M. Xu, S. Jia, Hyperspectral image denoising via robust subspace estimation and group sparsity constraint, *IEEE Trans. Geosci. Remote Sens.* 61 (2023) 1–16.
- [25] W. He, Q. Yao, C. Li, N. Yokoya, Q. Zhao, Non-local meets global: an integrated paradigm for hyperspectral denoising, in: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019, pp. 6861–6870.
- [26] Q. Zhang, Q. Yuan, M. Song, H. Yu, L. Zhang, Cooperated spectral low-rankness prior and deep spatial prior for HSI unsupervised denoising, *IEEE Trans. Image Process.* 31 (2022) 6356–6368.
- [27] Q. Zhang, Q. Yuan, Z. Li, F. Sun, L. Zhang, Combined deep prior with low-rank tensor SVD for thick cloud removal in multitemporal images, *ISPRS J. Photogramm. Remote Sens.* 177 (2021) 161–173.
- [28] L. Zhuang, X. Fu, M.K. Ng, J.M. Bioucas-Dias, Hyperspectral image denoising based on global and nonlocal low-rank factorizations, *IEEE Trans. Geosci. Remote Sens.* 59 (12) (2021) 10438–10454.
- [29] M. Maggioni, V. Katkovnik, K. Egiazarian, A. Foi, Nonlocal transform-domain filter for volumetric data denoising and reconstruction, *IEEE Trans. Image Process.* 22 (1) (2013) 119–133.
- [30] Y. Peng, D. Meng, Z. Xu, C. Gao, Y. Yang, B. Zhang, Decomposable nonlocal tensor dictionary learning for multispectral image denoising, in: 2014 IEEE Conference on Computer Vision and Pattern Recognition, 2014, pp. 2949–2956.
- [31] J. Xue, Y. Zhao, W. Liao, J.C.-W. Chan, Nonlocal low-rank regularized tensor decomposition for hyperspectral image denoising, *IEEE Trans. Geosci. Remote Sens.* 57 (7) (2019) 5174–5189.
- [32] P. Zhao, Q. Fan, R. Dang, J. Cao, Y. Zhang, Y. Feng, B. Hu, C. Gao, Q. Wang, Interference data denoising and baseline correction via a multi-scale deep denoising network and multidimensional gradient-consistent regularization, *Opt. Laser Technol.* 192 (2025) 113849.
- [33] X. Cao, X. Fu, C. Xu, D. Meng, Deep spatial-spectral global reasoning network for hyperspectral image denoising, *IEEE Trans. Geosci. Remote Sens.* 60 (2022) 1–14.
- [34] O. Sidorov, J.Y. Hardeberg, Deep hyperspectral prior: single-image denoising, inpainting, super-resolution, in: 2019 IEEE/CVF International Conference on Computer Vision Workshop (ICCVW), 2019, pp. 3844–3851.
- [35] Q. Zhang, Q. Yuan, J. Li, Z. Li, H. Shen, L. Zhang, Thick cloud and cloud shadow removal in multitemporal imagery using progressively spatio-temporal patch group deep learning, *ISPRS J. Photogramm. Remote Sens.* 162 (2020) 148–160.
- [36] X. Wang, R. Girshick, A. Gupta, K. He, Non-local neural networks, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2018, pp. 7794–7803.
- [37] F. Xiong, J. Zhou, Q. Zhao, J. Lu, Y. Qian, Mac-net: model-aided nonlocal neural network for hyperspectral image denoising, *IEEE Trans. Geosci. Remote Sens.* 60 (2022) 1–14.
- [38] Z. Fang, X. Ma, G.R. Arce, Degradation-estimated hybrid unfolding transformer network for efficient hyperspectral image reconstruction, *Opt. Laser Technol.* 192 (2025) 113338.
- [39] K. Wei, Y. Fu, H. Huang, 3-d quasi-recurrent neural network for hyperspectral image denoising, *IEEE Trans. Neural Netw. Learn. Syst.* 32 (1) (2021) 363–375.
- [40] E. Pan, Y. Ma, X. Mei, F. Fan, J. Huang, J. Ma, Sqad: spatial-spectral quasi-attention recurrent network for hyperspectral image denoising, *IEEE Trans. Geosci. Remote Sens.* 60 (2022) 1–14.
- [41] Z. Lai, Y. Fu, Mixed attention network for hyperspectral image denoising, *cs.CV*, arXiv:2301.11525, 2023.
- [42] Q. Zhang, Q. Yuan, J. Li, X. Liu, H. Shen, L. Zhang, Hybrid noise removal in hyperspectral imagery with a spatial-spectral gradient network, *IEEE Trans. Geosci. Remote Sens.* 57 (10) (2019) 7317–7329.
- [43] Y. Liu, R. Dian, S. Li, Low-rank transformer for high-resolution hyperspectral computational imaging, *Int. J. Comput. Vis.* 133 (2) (2025) 809–824.
- [44] Q. Zhang, Y. Dong, Y. Zheng, H. Yu, M. Song, L. Zhang, Q. Yuan, Three-dimension spatial-spectral attention transformer for hyperspectral image denoising, *IEEE Trans. Geosci. Remote Sens.* 62 (2024) 1–13.
- [45] Q. Zhang, J. Zhu, Y. Dong, E. Zhao, M. Song, Q. Yuan, 10-minute forest early wildfire detection: fusing multi-type and multi-source information via recursive transformer, *Neurocomputing* 616 (2025) 128963.
- [46] M. Li, J. Liu, Y. Fu, Y. Zhang, D. Dou, Spectral enhanced rectangle transformer for hyperspectral image denoising, in: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023, pp. 5805–5814.
- [47] Z. Lai, C. Yan, Y. Fu, Hybrid spectral denoising transformer with guided attention, *cs.CV*, arXiv:2303.09040, 2023.
- [48] S.W. Zamir, A. Arora, S. Khan, M. Hayat, F.S. Khan, M.-H. Yang, Restormer: Efficient transformer for high-resolution image restoration, *cs.CV*, arXiv:2111.09881, 2022.
- [49] Q. Zhang, X. Zhang, Y. Xiao, H. Xie, Deep low-rank tensor embedded network for hyperspectral image super-resolution, *Expert Syst. Appl.* 299 (2026) 129864.
- [50] Q. Zhang, Y. Zheng, Q. Yuan, M. Song, H. Yu, Y. Xiao, Hyperspectral image denoising: from model-driven, data-driven, to model-data-driven, *IEEE Trans. Neural Netw. Learn. Syst.* 35 (10) (2024) 13143–13163.
- [51] L. Pang, X. Rui, L. Cui, H. Wang, D. Meng, X. Cao, Hir-diff: Unsupervised hyperspectral image restoration via improved diffusion models, *arXiv preprint*, arXiv:2402.15865, 2024.
- [52] Y. Cai, J. Lin, Z. Lin, H. Wang, Y. Zhang, H. Pfister, R. Timofte, L.V. Gool, MST++: multi-scale spectral-wise transformer for efficient spectral reconstruction, in: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2022, pp. 744–754.
- [53] F. Xiong, J. Zhou, S. Tao, J. Lu, J. Zhou, Y. Qian, Smids-net: model guided spectral-spatial network for hyperspectral image denoising, *IEEE Trans. Image Process.* 31 (2022) 5469–5483.
- [54] Q. Fan, H. Huang, M. Chen, H. Liu, R. He, Rmt: retentive networks meet vision transformers, in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, pp. 5641–5651.
- [55] A. Katharopoulos, A. Vyas, N. Pappas, F. Fleuret, Transformers are RNNs: fast autoregressive transformers with linear attention, in: H.D. In: III, A. Singh (Eds.), *Proceedings of the 37th International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 119, 2020, pp. 5156–5165. PMLR.
- [56] S. Zhuoran, Z. Mingyuan, Z. Haiyu, Y. Shuai, L. Hongsheng, Efficient attention: attention with linear complexities, in: 2021 IEEE Winter Conference on Applications of Computer Vision (WACV), 2021, pp. 3530–3538.
- [57] S. Wang, B.Z. Li, M. Khabza, H. Fang, H. Ma, Linformer: Self-attention with linear complexity, *ArXiv*, arxiv:2006.04768, 2020.
- [58] K.M. Choromanski, V. Likhoshervostov, D. Dohan, X. Song, A. Gane, T. Sarlós, P. Hawkins, J.Q. Davis, A. Mohiuddin, L. Kaiser, D.B. Belanger, L.J. Colwell, A. Weller, Rethinking attention with performers, in: 9th International Conference on Learning Representations, ICLR 2021, Virtual Event, (3 May 2021), Austria, 2021.
- [59] A. Gu, T. Dao, Mamba: linear-time sequence modeling with selective state spaces, *ArXiv* (2024) 2312.00752 [cs.LG].
- [60] Y. Sun, L. Dong, S. Huang, S. Ma, Y. Xia, J. Xue, J. Wang, F. Wei, Retentive network: A successor to transformer for large language models, *ArXiv*, arxiv:2307.08621, 2023.
- [61] B. Arad, O. Ben-Shahar, Sparse recovery of hyperspectral signal from natural RGB images, in: European Conference on Computer Vision, 2016.
- [62] D.P. Kingma, J. Ba, Adam: A method for stochastic optimization, *CoRR*, arxiv:1412.6980, 2014.
- [63] L. Sun, C. He, Y. Zheng, Z. Wu, B. Jeon, Tensor cascaded-rank minimization in subspace: a unified regime for hyperspectral image low-level vision, *IEEE Trans. Image Process.* 32 (2023) 100–115.
- [64] L. Pang, W. Gu, X. Cao, Trq3dnet: a 3d quasi-recurrent and transformer based network for hyperspectral image denoising, *Remote. Sens.* 14 (2022) 4598.
- [65] W. Dong, H. Wang, F. Wu, G. Shi, X. Li, Deep spatial-spectral representation learning for hyperspectral image denoising, *IEEE Trans. Comput. Imaging* 5 (4) (2019) 635–648.